MODULE – VI COMPLETE ANSWERS (DBMS)

Based on the uploaded question bank.

SHORT ANSWER QUESTIONS (5 MARKS)

1. Explain Serial Schedule with Example

A **Serial Schedule** is a schedule in which transactions execute one after another without overlapping.

Example

Transactions:

T1:
Read(A)
Write(A)

T2:
Read(B)
Write(B)

Serial Schedule

T1: R(A) W(A)
T2: R(B) W(B)

or

T2: R(B) W(B)
T1: R(A) W(A)

Diagram

Time →

T1 : R(A) → W(A)
T2 :   R(B) → W(B)

Advantages

- Easy to understand
- No concurrency problems
- Always consistent

Disadvantages

- Poor resource utilization
- Low throughput

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2. Explain Non-Serial Schedule with Example

A **Non-Serial Schedule** allows operations of multiple transactions to be interleaved.

Example

T1: R(A)
T2: R(B)
T1: W(A)
T2: W(B)

Diagram

Time →

T1 : R(A) ----- W(A)
T2 :   R(B) ----- W(B)

Advantages

- Better CPU utilization
- Faster execution
- Supports concurrency

Disadvantages

- Can cause anomalies
- Requires concurrency control

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3. Explain Transaction Control Language (TCL)

TCL commands manage database transactions.

COMMIT

Makes changes permanent.

```
COMMIT;
```

Example:

```
UPDATE Employee
SET Salary=50000
WHERE EmpID=1;

COMMIT;
```

ROLLBACK

Undoes changes since last commit.

```
ROLLBACK;
```

SAVEPOINT

Creates a checkpoint.

```
SAVEPOINT S1;
```

ROLLBACK TO SAVEPOINT

ROLLBACK TO S1;

TCL Commands Summary

Command	Purpose
COMMIT	Save changes
ROLLBACK	Undo changes
SAVEPOINT	Create checkpoint
ROLLBACK TO	Return to checkpoint

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4. Define Lock. Discuss Two-Phase Locking Protocol (2PL)

Lock

A lock is a mechanism used to control concurrent access to database items.

Purpose

- Prevent inconsistency
- Maintain isolation
- Avoid conflicts

Types of Locks

Shared Lock (S)

Allows reading only.

Multiple transactions can read.

Exclusive Lock (X)

Allows read and write.

Only one transaction allowed.

Two-Phase Locking Protocol (2PL)

Transaction executes in two phases.

Phase 1: Growing Phase

Locks can be acquired.

Locks cannot be released.

Acquire Lock
Acquire Lock
Acquire Lock

Phase 2: Shrinking Phase

Locks can be released.

No new locks can be acquired.

Release Lock
Release Lock
Release Lock

Diagram

Lock Point
▲

```
Growing       |      Shrinking
Phase        |      Phase

Lock Lock Lock|Unlock Unlock
-------------+---------------
```

Advantages

- Ensures conflict serializability
- Maintains consistency

Disadvantages

- Deadlock may occur

---

5. When is a Schedule Called Non-Recoverable Schedule?

A schedule is **non-recoverable** if a transaction commits before the transaction from which it read data commits.

Example

T1: W(A)
T2: R(A)
T2: COMMIT
T1: ABORT

Problem

T2 committed using uncommitted data.

After T1 aborts:

Database becomes inconsistent.

Conclusion

Such schedules should be avoided.

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6. Difference Between Serial Schedule and Serializable Schedule. Explain Types of Locks

Serial Schedule

Transactions execute one after another.

Example:

T1 complete
T2 complete

Serializable Schedule

Transactions may interleave but produce same result as a serial schedule.

Comparison

Serial	Serializable
No overlap	Overlap allowed
Simple	Complex
Low concurrency	High concurrency
Always correct	Correct if serializable

Types of Locks

Shared Lock (S)

Read only.

Example:

LOCK-S(A)
READ(A)

Exclusive Lock (X)

Read and write.

Example:

LOCK-X(A)
WRITE(A)

Binary Lock

Only two states:

Locked
Unlocked

Intention Locks

Used in multi-level locking.

Examples:

IS
IX
SIX

LONG ANSWER QUESTIONS (10 MARKS)

1. Explain ACID Properties of Transaction with Example

A transaction must satisfy ACID properties.

A – Atomicity

Transaction executes completely or not at all.

Example

Transfer ₹1000:

A Account -1000
B Account +1000

Both operations must occur.

C – Consistency

Database remains valid before and after transaction.

Example

Total balance remains unchanged.

I – Isolation

Transactions should not interfere.

Example

Two users updating same account simultaneously.

D – Durability

Committed changes remain permanent.

Example

Even after power failure.

Diagram

```
Transaction
     |
-------------------------
|     |      |         |
A     C      I         D
```

Banking Example

Before Transfer

A = 5000
B = 3000

After Transfer

A = 4000
B = 4000

Total:

8000

Consistency maintained.

---

2. Discuss States of Transaction with Suitable Diagram

A transaction passes through several states.

Active State

Transaction executing.

Partially Committed State

Last statement executed.

Committed State

Changes permanently saved.

Failed State

Error occurs.

Aborted State

Rollback performed.

Terminated State

Transaction ends.

Diagram

```
          Active
             |
             ▼
   Partially Committed
             |
     ----------------
     |            |
     ▼            ▼
  Committed      Failed
     |            |
     ▼            ▼
 Terminated     Aborted
                  |
                  ▼
              Terminated
```

Explanation

1. Active → executing.
2. Partially Committed → last operation done.
3. Committed → COMMIT executed.
4. Failed → error detected.
5. Aborted → rollback performed.
6. Terminated → transaction completed.

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3. Explain Lock-Based Protocols in Detail

Lock-based protocols use locks to control concurrent transactions.

Shared Lock

Used for reading.

LOCK-S(A)
READ(A)

Multiple transactions allowed.

Exclusive Lock

Used for writing.

LOCK-X(A)
WRITE(A)

Only one transaction allowed.

Two-Phase Locking (2PL)

Growing Phase

Acquire locks only.

Shrinking Phase

Release locks only.

Example

T1:

Lock-X(A)
Read(A)
Write(A)

Lock-X(B)
Read(B)
Write(B)

Unlock(A)
Unlock(B)

Strict 2PL

Locks released only after commit.

Advantages:

- Prevents cascading rollback.
- Ensures recoverability.

Rigorous 2PL

All locks released after commit.

Most restrictive.

Advantages

- Consistency
- Serializability
- Isolation

Disadvantages

- Deadlock
- Starvation

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4. Precedence Graph and Conflict Serializability

Given Schedule

S:
r1(x)
r2(y)
r3(y)
w2(y)
w1(x)
w3(x)
r2(x)
w2(x)

Step 1: Identify Conflicts

On X

w1(x) → w3(x)

Edge:

T1 → T3

---

w1(x) → r2(x)

Edge:

T1 → T2

---

w1(x) → w2(x)

Edge:

T1 → T2

---

w3(x) → r2(x)

Edge:

T3 → T2

---

w3(x) → w2(x)

Edge:

T3 → T2

On Y

r3(y) → w2(y)

Edge:

T3 → T2

Step 2: Construct Precedence Graph

```
T1 ──────▶ T3
 |      |
 |      ▼
 └───────▶ T2
```

Step 3: Check Cycle

Graph contains:

T1 → T3
T3 → T2
T1 → T2

No cycle exists.

Step 4: Conclusion

Since graph is acyclic:

Schedule is Conflict Serializable

Serialization Order

T1 → T3 → T2
